

tests/final-verify.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Authority:** Helix Constitution §11.4.1 (no false-FAIL), §11.4.68 (no fail-open), §11.4.3 (topology SKIP), §11.4.69 (data-plane egress proof), discovery-sweep F3 / BUGFIX-0014

Companion (§11.4.18) to the in-source comment blocks in the script.

Overview / Purpose

A **live final-verification smoke check** of the running proxy service: it drives four through-proxy connectivity checks plus a VPN-routing proof and prints a banner-boxed pass/fail/skip summary. It is functionally the same connectivity check as tests/verify-proxy.sh (identical conn_check + VPN logic) with a boxed "FINAL VERIFICATION" presentation. Each connectivity check is anti-bluff: a proxy miss is classified via proxy_conn_verdict into PASS/FAIL/SKIP rather than blindly failing on a third-party outage.

Usage

```
bash tests/final-verify.sh
```

No CLI flags or arguments.

Inputs

- **.env** (optional, gitignored) — sourced if present for HTTP_PROXY_PORT (default 53128), SOCKS_PROXY_PORT (default 51080), and VPN_EXIT_IP.
- **VPN_EXIT_IP** (optional) — the expected VPN tunnel exit IP. Unset → the VPN-routing check SKIPS (operator_attended, live RED/GREEN deferred to P10); it never fabricates a VPN PASS from an egress == host equality.
- Live endpoints reached (through the proxy, and directly for cross-checks): connectivitycheck.gstatic.com/generate_204,

www.google.com, and ifconfig.me (host-IP probe when VPN_EXIT_IP is set).

Outputs

- Boxed banner + one line per check (HTTP proxy, HTTPS via HTTP proxy, SOCKS5 proxy, HTTPS via SOCKS5, VPN routing) + a SUMMARY: Passed/Failed/Skipped box.
- Exit 0 — FAIL == 0 (skips allowed); exit 1 — one or more FAIL.

Side-effects

None to the tree/containers. It issues real curl requests through the proxy and (for the anti-bluff cross-check) directly. It does not start/stop/reconfigure anything.

Dependencies

- bash (set -euo pipefail), curl.
- tests/lib/evidence.sh (sourced) — _code_in, port_is_listening, proxy_conn_verdict, assert_egress_ip, ab_skip_with_reason.

Internal behaviour notes

- test_pass/test_fail use assignment-form counters (§11.4.1 — ((PASS++)) from 0 aborts under set -e before the VPN check).
- conn_check: proxy returns an expected code → PASS; proxy up but cannot fetch a directly-reachable site → FAIL (§11.4.68 no fail-open); proxy+direct both fail → SKIP (§11.4.1 no false-FAIL on an outage); port not listening → SKIP.

Related scripts

- tests/verify-proxy.sh — the near-identical lighter-presentation sibling (same conn_check + VPN logic).
- tests/comprehensive-test.sh — the full functional suite this smoke check distils.
- tests/regression/proxy_conn_verdict_test.sh — the §11.4.135 guard for the proxy_conn_verdict classifier both scripts use.

Last verified

2026-07-01 — documented from source. Behaviour read from the script body; not executed here (the conductor runs the live check against the running stack).